[Aptitude Multiple Choice Question – Rahul Kale (rahul@sunbeaminfo.com)] [9881144174]

[Percentage] Q1. In an examination 80% candidates passed in English and 85% candidates passed in Mathematics. If 73% candidates passed in both these subjects, then what per cent of candidates failed in both the subjects?

A. 8 B. 15 C. 27 D. 35

Students passed in English = 80%

Students passed in Math's = 85%

Students passed in both subjects = 73%

Then, number of students passed in at least one subject .

= (80 + 85) - 73 = 165 - 73

= 92%.

The percentage of students passed in English and Maths individually, have already included the percentage of students passed in both subjects. So, We are subtracting percentage of students who have passed in both subjects to find out percentage of students at least passed in one subject

Thus, students failed in both subjects = 100-92 = 8%.

Answer : A is correct 8

[Percentage] Q2. The price of the sugar rise by 25%. If a family wants to keep their expenses on sugar the same as earlier, the family will have to decrease its consumption of sugar by

A. 20 % **B. 25 %** **C. 80 %** **D. 75 %**

Let the initial expenses on Sugar was Rs. 100.

Now, Price of Sugar rises 25%. So, to buy same amount of Sugar, they need to expense,

= (100 + 25% of 100) = Rs. 125.

But, They want to keep expenses on Sugar, so they have to cut Rs. 25 in the expenses to keep it to Rs. 100.

Now, % decrease in Consumption,

25 / 125 * 100 = 1/5 *100 = 20%

Another Calculation Method; 100 === 25%↑ ===> → 125 === X%↓ ===> 100

X = 25 / 125 * 100 = 20%

Answer : A is correct 20 %

Q3. Population of a town increase 2.5% annually but is decreased by 0.5% every year due to migration. What will be the percentage increase in 2 years?

A. 5 % B. 4:04 % C. 4 % D. 3:96 %

Net percentage increase in Population = (2.5 - 0.5) = 2% each year.

Let the Original Population of the town be 100.

Population of Town after 1 year = (100 + 2 % of 100) = 102.

Population of the town after 2nd year = (102 + 2 % of 102) = 104.04

Now, % increase in population = 4.04 / 100 * 100 = 4.04%

Another Calculation Method:

100 == 2 % Up (1st year) ==> 102 == 2 % Up (2nd year) ==> 104.04

% population increase in 2 years = 4.04%.

Answer : B is correct 4:04 %

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Q4. In an election between two candidates, the winner got 65% of the total votes cast and won the election by a majority of 2748 votes. What is the total number of votes cast if no vote is declared invalid?

A. 8580 B. 8720 C. 9000 D. 9160

Winner gets 65% of valid votes and loser gets 35% of votes

Difference between this two = 2748

(65 – 35) % = 2748

(30) % = 2748

Total number of voters, 100%

= 2748 * 100 / 30

= 9160

Answer : D is correct 9160

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[Percentage] Q5. Amol spends 30% of his income on education and 50% of the remaining on food. He gives Rs. 1000 as monthly rent and now has Rs. 1800 left with him. What is his monthly income?

A. Rs. 8000 B. Rs. 7000 C. Rs. 9000 D. Rs. 6000

Amol's saving and rent = 1000 + 1800 = Rs. 2800

Let his monthly income be Rs. 100. 30% of his income he spent on education i.e. Rs. 30

Remaining = 100 - 30 = 70

50% of remaining on food = 70 * 50 / 100 = 3500 / 100 = rs. 35

Now, that 35 must be equal to his saving and rent i.e . 35 = 2800 then,

1 = 2800 / 35

100 = 2800 * 100 / 35 = 2800 / 35 * 100 = 8000

Rs. 8000

Answer : A is correct 8000

[Ratio] Q6. The number of students in 3 classes is in the ratio 2 : 3 : 4. If 12 students are increased in each class this ratio changes to 8 : 11 : 14. The total number of students in the three classes in the beginning was ?

A. 162 B. 108 C. 96 D. 54

Let the number of students in the classes be 2x, 3x and 4x respectively;

Total students = 2x + 3x + 4x = 9x

According to the question , 2x +12 / 3x+12 = 8 / 11

8 * (3x+ 12) = 11 * (2x+12)

24x + 96 = 22x +132

24x - 22x = 132 -96

2x=132−96 2x= 36 x=18

Hence , Original number of students, 9x = 9 × 18 = 162

Answer : A is correct 162

[Ratio] Q7. The salaries of A, B and C are in the ratio 1 : 3 : 4. If the salaries are increased by 5%, 10% and 15% respectively, then the increased salaries will be in the ratio?

A. 20 : 66 : 95 B. 21 : 66 : 96 C. 21 : 66 : 92 D. 19 : 66 : 92

Let A's Salary = Rs. 100

Then, B's Salary = Rs. 300

And, C's Salary = Rs. 400 here Salary has given in 1 : 3 : 4 ratio

Now, 5% increase in A's Salary, A's new Salary = (100 + 5% of 100) = Rs. 105

B's Salary increases by 10%, Then, B's new Salary = (300 + 10% of 300) = Rs. 330

C's Salary increases by 15%,C's new Salary = (400 + 15% of 400) = Rs. 460

Then, ratio of increased Salary, A : B : C = 105 : 330 : 460

A : B : C = 105/5 : 330 /5: 460/5 = 21 : 66 : 92

Answer : C is correct 21 : 66 : 92

Q8. In a school having roll strength 286, the ratio of boys and girls is 8 : 5. If 22 more girls get admitted into the school, the ratio of boys and girls becomes

A. 12 : 7 B. 10 : 7 C. 8 : 7 D. 4 : 3

Boys : girls = 8 : 5 (let the boys = 8x, girl = 5x)
Total strength = 286
8x + 5x = 286

13x = 286 x= 286 / 13 = 22

Boys= 22*8 =176 Girls= 22*5=110

Boys = 176 and girls = 110

22 more girls get admitted then number of girls become

(5x + 22) = 110 + 22 = 132

Now, new ratio of boys and girls = 176 : 132 = 176/44 : 132/44 4 : 3

Answer : D is correct 4 : 3

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[Alligation] Q9. In a 729 litres mixture of milk and water, the ratio of milk to water is 7 : 2. to get a new mixture containing milk and water in the ratio 7 : 3, the amount of water to be added is

A. 81 litres B. 71 litres C. 56 litres D. 50 litres

Quantity of milk in 729 liter of mixture = (7 * 729) / 9 == 567

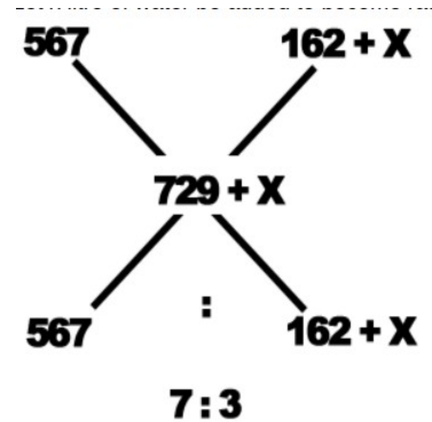
Quantity of water = 729- 567 = 162 liter

Let x litre of water be added to become ratio 7 : 3

567 162 +x

729+x 7:3

567 162+x



7 / 3 = 567 / (162+x)

162 * 7 + 7x = 567 *3

1134 + 7x = 1701

7x = 1701 – 1134 = 567

x = 567 / 7 = 81 liter

If we add 81 litres water ratio will be 7: 3

Answer : A is correct 81 litres

[Alligation] Q10. The ratio, in which tea costing Rs. 192 per kg is to be mixed with tea costing Rs. 150 per kg so that the mixed tea when sold for Rs. 194.40 per kg, gives a profit of 20%

A. 2 : 5 B. 3 : 5 C. 5 : 3 D. 5 : 2

Cost Price of first tea = Rs. 192 per kg.

Cost Price of Second tea = Rs. 150 per kg.

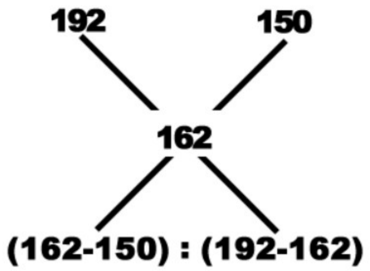
Mixture is to be sold in Rs. 194.40 per kg, which has included 20% profit. So,

Selling price of Mixture = Rs. 194.40 per kg.

Let the CP of Mixture be Rs. X per kg. Therefore, X + 20% of X = SP

6x / 5 =194.40 6x= 194.40 * 5 6x= 972 x= 972/6 x= 162

Let N kg of first tea and M kg of second tea to be added.



162 150 : 192- 162

N / M = 12 / 30 (12/6) / (30 / 6) ---> 2 : 5

Answer : A is correct 2: 5

[Alligation] Q10. The ratio, in which tea costing Rs. 192 per kg is to be mixed with tea costing Rs. 150 per kg so that the mixed tea when sold for Rs. 194.40 per kg, gives a profit of 20% (2ed method)

A. 2 : 5 **B. 3 : 5** **C. 5 : 3** **D. 5 : 2**

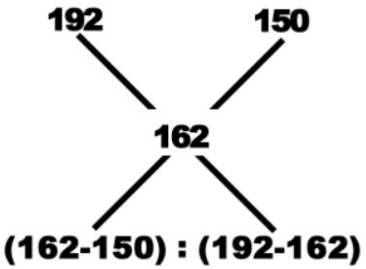
Cost Price of first tea = Rs. 192 per kg.

Cost Price of Second tea = Rs. 150 per kg.

Mixture is to be sold in Rs. 194.40 per kg, which has included 20% profit. So,

Selling price of Mixture = Rs. 194.40 per kg.

Selling price of Mixture =Cost Price *(100+ gain %) / 100



194.40 = Cost Price * (100 + 20) / 100

194.40 = Cost Price * (120) / 100

19440 = Cost Price * 120

Cost Price =19440 / 120 = 162 **12 : 30** **2 : 5**

Answer : A is correct 2: 5

Q11. In a zoo, there are Rabbits and Pigeons. If heads are counted, there are 200 and if legs are counted, there are 580. How many pigeons are there?

A. 90 B. 100 C. 110 D. 120

Rabbits has 4 legs and Pigeons has 2 legs

let x will be number of rabbits and y will be number of pigeons.

4x+2y= 580 2x +y =290 ---> equation 1

x+y=200

y= 200-x substitute y in equation 1 2x +y =290

2 x + (200 -x) =290

x +200 = 290 ie x= 90

So x + y = 200 y=200-90 ==110

Answer : C is correct 110

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Q12. In what ratio must a mixture of 30% alcohol strength be mixed with that of 50% alcohol strength so as to get a mixture of 45% alcohol strength?

A. 1 : 2   B. 1 : 3   C. 2 : 1   D. 3 : 1

Let the ratio be x:y

Then according to the question

$30x + 50y = (x+y)*45$

$30x + 50y = 45x + 45y$

$50y – 45y = 45x - 30x$

$5y = 15x$ $x/y = 5/15$ 1 : 3

Ratio 1:3

Answer : B is correct 1 : 3

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Q13. A mixture of 150 liters of spirit and water contains 20% water. How much more water should be added so that water becomes 25% of the new mixture?

A. 5 lit B. 20 lit C. 15 lit D. 10 lit

Volume of water in mixture = 20% of 150 = 30 liters.

Volume of spirit in mixture = 150 -30 =120 liters

y' more litres of water added to mixture

New total volume of mixture = 150 + y

New Mixture, Water = 25% and Spirit = 75% **75% of (150 + y) = 120**

75 /100 * (150 + y) = 120

75 * (150 + y) = 120 * 100

75*150 + 75y = 12000
11250 + 75y = 12000 **75y = 12000 -11250**

75y =750 **y=750/75** **y=10**

Answer : D is correct 10 lit

[Speed, Time, and Distance]   Q14. A boy goes to school from home at a speed of 10km/hr and return back at 30km/hr. Find his average speed.

A. 15 km/hr   B. 14.5 km/hr   C. 10 km/hr   D. 20 km/hr

average speed = 2ab / (a+b)

a = 10 km/hr

b= 30 km/hr

average speed =   2*10*30 / (10+30)

average speed =   20*30 / (10+30)

average speed =   600 / 40

average speed =   60   / 4

average speed =   15 km/hr

Answer : A   is correct 15 km/hr

Q15. On a journey, across Delhi, a Taxi averages 30 kmph for 60% of the distance, 20 kmph for 20% of it and 10kmph for the remainder. The average speed for the whole journey is : way 1

A. 20 km/hr B. 22.5 km/hr C. 24.625 km/hr D. 25 km/hr

Let total journey = y km

Total time taken = time taken during 60% of distance + time taken during 20% of distance + time taken in remaining distance

= [(60y / 100) / 30] + [(20y / 100) / 20] + [(20y / 100) / 10]

= (y / 50 + y / 100 + y / 50) hrs

= (2y / 100 + y / 100 + 2y / 100) hrs

= (5y/100) hrs

= y/20 hours.

Average speed = total distance traveled / total time taken.

= [y / (y/20)] km/hr
= (20/y * y / (y/20) *20/y)
= 20 /1 / 1

Answer : A is correct 20 km/hr

Q15. On a journey, across Delhi, a Taxi averages 30 kmph for 60% of the distance, 20 kmph for 20% of it and 10kmph for the remainder. The average speed for the whole journey is : **way 2**

A. 20 km/hr **B. 22.5 km/hr** **C. 24.625 km/hr** **D. 25 km/hr**

30 km / hr	20 km / hr	10 km / hr	
			rest
60%	20%	rest	100- 60-20

Average speed = total distance traveled / total time taken.

Time= distance / speed

avg speed = d1 + d2 + d3 / t1 + t2 + t3

= d1 + d2 + d3 / (d1 / s1) + (d2 / s2) + (d3 / s3)

= 60 + 20 + 20 / (60 / 30) + (20 / 20) + (20/10)

= 60 + 20 + 20 / (2) + (1) + (2)

= 100 / 5

= 20 km / hr

Answer : A is correct 20 km/hr

Q16. A distance is covered by a cyclist at a certain speed. If a jogger covers half of the
distance in double the time, the ratio of the speed of the jogger to that of the cyclist is :

A. 1 : 4 B. 4 : 1 C. 1 : 2 D. 2 : 1

Let the distance covered by the cyclist be x and the time taken be y

Then, required ratio : = 1/2x / 2y

Express as single fraction

= 1/ 2x*2y

Multiply 2 and 2 to get 4.

= 1/4xy

: x/y = 1/4

So ratio 1 : 4

Answer : A is correct 1 : 4

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Q17. A and B together can do a piece of work in 8 days. If A alone can do the same work in 12 days. A alone worked for 4 days. After that how long will B alone take to complete the work?

A. 20 days **B. 16 days** **C. 15 days** **D. 18 days**

LCM (8, 12) = 24

One day work of A + B = 24 / 8 = 3 unit / day

A's 1 day work = 24/ 12 = 2 units / days

B's 1 day work = 3 - 2 = 1 units/days

4 days work ofA = 4 × 2 = 8 units/days

Work left = 24 - 8 = 16 units

B will complete the remaining work in = 16 units / 1 unit per day =16 days

Answer : B is correct 16 Days

Q18. The average weight of 10 men is decreased by 2 kg when one of them whose weight is 60 kg is replaced by a new man. What is the weight of the new man?

A. 35 kg B. 40 kg C. 45 kg D. 50 kg

Average decrease in weight per person = 2kg

There are 10 men so total decrease in weight on replacing a man with a new man = 10 * 2= 20

Weight of the man who is replaced =60 kg

The weight of the new man would be = Weight of man replaced - total decrease in weight of the group of 10 men.

The weight of the new man would be] = 60 - 20

The weight of the new man would be = 40

Answer : B is correct 40 Kg